



Object Oriented Programming Lab

Course Code: CSE 1116

Home Assignment – 3

[Q1] An online shopping platform needs a system to manage its products. You are required to create a package named store to keep your classes organized. At the core of the program, design a base class called Product with fields for product ID, name, and price. The store sells different types of items, so you must create subclasses such as Electronics (with warranty period), Clothing (with size), and Book (with author).

The system should use polymorphism so that when displayDetails() is called, each product type shows its own specific details. All the products must be stored inside an ArrayList<Product> for easy management.

Next, write a **menu-based program** that allows the user to:

1. Add a new product (with a choice for Electronics, Clothing, or Book).
2. Display all products in the store.
3. Search for a product by name.
4. Calculate the total price of all products currently in the store.
5. Exit the program.

Look for the sample output bellow;

Program Console

```
===== Online Shopping Store =====
1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit
Enter your choice: 1

Enter Product Type (1. Electronics, 2. Clothing, 3. Book): 1
Enter Product ID: P101
Enter Product Name: Laptop
Enter Product Price: 85000
Enter Warranty Period (in years): 2
Product added successfully!

===== Online Shopping Store =====
1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit
Enter your choice: 1

Enter Product Type (1. Electronics, 2. Clothing, 3. Book): 2
Enter Product ID: P102
Enter Product Name: T-Shirt
Enter Product Price: 1200
Enter Size: M
Product added successfully!
```

===== Online Shopping Store =====

1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit

Enter your choice: 1

Enter Product Type (1. Electronics, 2. Clothing, 3. Book): 3

Enter Product ID: P103

Enter Product Name: Java Programming

Enter Product Price: 500

Enter Author: James Gosling

Product added successfully!

===== Online Shopping Store =====

1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit

Enter your choice: 2

--- All Products in Store ---

[Electronics] ID: P101, Name: Laptop, Price: 85000, Warranty: 2 years

[Clothing] ID: P102, Name: T-Shirt, Price: 1200, Size: M

[Book] ID: P103, Name: Java Programming, Price: 500, Author: James Gosling

===== Online Shopping Store =====

1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit

Enter your choice: 3

Enter product name to search: T-Shirt

Product Found: [Clothing] ID: P102, Name: T-Shirt, Price: 1200, Size: M

===== Online Shopping Store =====

1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit

Enter your choice: 4

Total Price of All Products = 86700

===== Online Shopping Store =====

1. Add Product
2. Display All Products
3. Search Product by Name
4. Calculate Total Price
5. Exit

Enter your choice: 5

Exiting... Thank you for using Online Shopping Store!